

ARCANE CRYSTALS (LITE)

Technical Integration Manual & Official Readme

Thank you for choosing **Arcane Crystals**! This is an introductory **Lite Starter Pack** designed to give you a taste and preview of the full, expanded version. Every single mesh is modeled and optimized to ensure maximum performance in your game scenes, regardless of your target platform.



Figure 1: Visual preview of the crystal formations included in this Lite edition.

1. Package Contents

The Lite package offers 5 distinct organic shapes to get you started, pre-configured across 4 vibrant color palettes.

Feature	Technical Asset Details
5 Unique Shapes	A curated selection of distinct crystal formations designed for quick environment decoration and testing.
4 Colors Available	Cyan, Green, Purple, and Red.
Pipeline Ready	Built-In, URP, and HDRP versions are all fully supported and available out of the box.
Single Texture Atlas	All models and color variants share a single material texture atlas. Unity batches thousands of meshes into **exactly 1 Draw Call** using GPU Instancing.

 **Baked Vertex Color Ambient Occlusion:** I integrated ambient occlusion values directly into the **Vertex Color data** of the meshes. This preserves deep, rich geometric shadowing even if you choose to use lightweight unlit or custom stylized shaders!

2. Material Setup & Render Pipelines

All 3 render pipelines (**Built-In**, **URP**, and **HDRP**) are available and pre-configured. Select the setup that matches your active Unity graphics settings.

Universal RP (URP)

- **Default Setup:** Ready to use immediately.
- All prefabs are pre-configured with URP-ready materials and shaders.
- Drag and drop the prefabs directly into your scene.

High Definition RP (HDRP)

- **Materials Included:** Dedicated HDRP materials are available inside the package.
- Assign the provided HDRP materials to the crystal meshes in your scene.
- *(See below to import HDRP to your project).*

Built-In / Standard

- **Materials Included:** Standard pipeline materials are provided in the project folder.
- Assign these materials using Unity's default Standard Shader.
- Remove any URP-specific scriptable settings files if any console warnings appear.

How to install HDRP if it is not present in your Unity Project:

1. In the top menu, go to **Window > Package Manager**.
2. In the top-left dropdown of the Package Manager window, select **Packages: Unity Registry**.
3. Type **High Definition RP** in the search bar.
4. Select it from the list and click the **Install** button in the bottom right.
5. Once installed, the pre-configured HDRP materials inside the package will immediately render correctly on your models.